



# Researchers, come investigate!

We will solve a model sand case, collect real evidence, and save one question for camp.

CASE



PREDICT



LOOK



MAGNIFY



MATCH



REFLECT



SAMPLE



WONDER

**You do not have to hold the whole lesson in your head. We will show one step at a time.**



Point



Draw or write



Speak or dictate



Photograph



Work with a partner



Choose a seated role



Ask for help



Pass and rejoin



# Enter the fictional case

This story lets us practice evidence matching before we collect real Charlotte Beach sand.



## Durand reference model

Prepared model, not beach data



## Ontario reference model

Prepared model, not beach data



## Mystery bird clue

Built to match one reference

**MODEL STORY:** No real injured bird is being reported. A match cannot tell us why an animal needed care.

**Which prepared reference model best matches the mystery clue?**



# Predict

Make a first idea before gathering more evidence. A prediction is allowed to change.

**Which reference model will have more visible proxy pieces?**



**Durand model**



**About the same**



**Ontario model**

**Choose a response route:** point, circle, speak, write, draw, tell a partner, or ask us to record your idea.

**Good science is not protecting a first guess.** Good science is revising when the evidence gives us a better explanation.



# Look first

Use the same method with each prepared reference model.



## Read the label.

Keep each model separate.



## Work over a tray.

Keep proxy pieces contained.



## Look without tools.

Notice color, shape, and grain size.



## Count what you see.

Record the first-look count.



## Record a detail.

Observation first; meaning later.



## Pause if needed.

Your partner can hold the place.

## What might a first look miss?



# Magnify and look again

Changing the tool may change what becomes visible; it does not change the evidence standard.



## Use one magnifier.

Move the tool or tray slowly.



## Search the full sample.

Use a repeatable pattern.



## Count again.

Record the magnified count.



## Compare counts.

What stayed the same?



## Name what changed.

Use a specific observation.



## Return every piece.

Reset the model over the tray.

**I noticed ...**

**I think this could mean ...**

**We still need to know ...**



# Match, explain, and revise

The best match uses more than one feature. The model has limits.

## Which reference best matches the mystery clue?

### Use two details

Compare grain color or size, proxy color or shape, first-look count, magnified count, or another repeatable pattern.

**My claim is ...**

**The evidence is ...**

### Name one limit

The prepared model can support a known match. It cannot measure either beach, trace a real animal, confirm plastic, or prove a health effect.

**Revising is evidence of learning.** A changed answer is welcome when the reason is clear.

You may explain by pointing to features, drawing, speaking, writing, using the sentence starters, or asking a partner or scribe to share your evidence.



# Collect one real sand sample

Every pair follows the same method so six locations remain traceable and comparable.



## 1. CHOOSE

Go to the assigned safe site.



## 2. FRAME

Place the red square flat.



## 3. SCOOP

Collect the top 1/2 inch.



## 4. SEAL

Close the jar before moving.



## 5. LABEL

Pair + site + time.



## 6. MAP

Draw or mark the site.

**STOP:** Do not add salt water at the beach. Do not combine samples. Do not call a visible field fragment confirmed plastic.



# Save one question for Monday

The prepared model had a known answer. The real beach sample does not.

## What do you notice, think, or still need to know?

### Possible questions

What might a first look miss?  
How could plastic move through connected systems?  
What evidence could confirm a suspected particle?  
Where could a fair system change interrupt the pathway?

### Choose a route

Write, draw, dictate, photograph your card, tell a partner, place a sticky note, or pass and return later.

SAMPLE



ADD BRINE



SETTLE



FILTER

**For Monday:** Every field sample stays sealed, separate, and connected to its own ID, site record, filter, image, and observation.



# Durand model evidence record

Practice model only. This is not a measurement of Durand Eastman Beach.

Name(s) \_\_\_\_\_ Rotation \_\_\_\_\_ Date \_\_\_\_\_

## SCIENCE | BEFORE YOU LOOK

Which prepared model do you predict will contain more visible proxy pieces?

☐ Durand ☐ Ontario ☐ About the same

What makes you think that?

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Confidence: ☐ Not yet ☐ A little ☐ Mostly ☐ Very

## EVIDENCE | FIRST LOOK

Visible proxy pieces counted \_\_\_\_\_

I noticed ...

---



---



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This step felt:

☐ Easy ☐ Okay ☐ Tricky ☐ I need another route

## EVIDENCE | MAGNIFIED LOOK

Visible proxy pieces counted \_\_\_\_\_

What became easier to notice, or what changed?

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---



---

This step felt:

☐ Easy ☐ Okay ☐ Tricky ☐ I need another route

## EVIDENCE | CAREFUL CLAIM LANGUAGE

I noticed

I think this could mean

We still need to know

---

**Choose your route:** write, draw, point, dictate, work with a partner, or ask for a scribe.

**Evidence limit:** This prepared jar helps us practice comparison. A visible piece in a real sample is only *suspected* plastic until appropriate analysis confirms the material.



# Ontario model evidence record

Practice model only. This is not a measurement of Ontario Beach Park.

Name(s) \_\_\_\_\_ Rotation \_\_\_\_\_ Date \_\_\_\_\_

## SCIENCE | BEFORE YOU LOOK

Which prepared model do you predict will contain more visible proxy pieces?

☐ Durand ☐ Ontario ☐ About the same

What makes you think that?

---



---

Confidence: ☐ Not yet ☐ A little ☐ Mostly ☐ Very

## EVIDENCE | FIRST LOOK

Visible proxy pieces counted \_\_\_\_\_

I noticed ...

---



---



---

This step felt:

☐ Easy ☐ Okay ☐ Tricky ☐ I need another route

## EVIDENCE | MAGNIFIED LOOK

Visible proxy pieces counted \_\_\_\_\_

What became easier to notice, or what changed?

---



---



---

This step felt:

☐ Easy ☐ Okay ☐ Tricky ☐ I need another route

## EVIDENCE | CAREFUL CLAIM LANGUAGE

I noticed

I think this could mean

We still need to know

---

**Choose your route:** write, draw, point, dictate, work with a partner, or ask for a scribe.

**Evidence limit:** This prepared jar helps us practice comparison. A visible piece in a real sample is only *suspected* plastic until appropriate analysis confirms the material.



# Investigate the sand

Fictional model case | Card 1

**MODEL STORY:** No real sick bird is being reported. The case cannot show why an animal needed care.

**YOUR QUESTION:** Which prepared reference model best matches the mystery clue?



## PREDICT

Make a first idea.



## LOOK

Count what is visible.



## MAGNIFY

Count again.



## MATCH

Use two details.

**Investigators revise.** You do not need a perfect first answer. Change your idea when evidence gives you a better explanation.



**Point, draw, speak, write, photograph, use a partner or scribe, or pass and rejoin.**

CUT HERE



# Investigate the sand

Fictional model case | Card 2

**MODEL STORY:** No real sick bird is being reported. The case cannot show why an animal needed care.

**YOUR QUESTION:** Which prepared reference model best matches the mystery clue?



## PREDICT

Make a first idea.



## LOOK

Count what is visible.



## MAGNIFY

Count again.



## MATCH

Use two details.

**Investigators revise.** You do not need a perfect first answer. Change your idea when evidence gives you a better explanation.



**Point, draw, speak, write, photograph, use a partner or scribe, or pass and rejoin.**



## Case reflection

Use evidence; revision is welcome | Card 1

Name(s) \_\_\_\_\_ Rotation \_\_\_\_\_

### 💡 AT FIRST

I predicted

because

### ⚖️ AFTER INVESTIGATING

The best match is

Evidence 1

Evidence 2

### ⚠️ MODEL LIMIT

This model cannot tell us

### ❓ I STILL WONDER

✍️ Write, draw, point, dictate, work with a partner, or ask for a scribe.

CUT HERE



## Case reflection

Use evidence; revision is welcome | Card 2

Name(s) \_\_\_\_\_ Rotation \_\_\_\_\_

### 💡 AT FIRST

I predicted

because

### ⚖️ AFTER INVESTIGATING

The best match is

Evidence 1

Evidence 2

### ⚠️ MODEL LIMIT

This model cannot tell us

### ❓ I STILL WONDER

✍️ Write, draw, point, dictate, work with a partner, or ask for a scribe.



# Charlotte Beach sample record

Real field sample | Card 1

Pair \_\_\_\_\_ Sample ID \_\_\_\_\_ Time \_\_\_\_\_

Site or nearby landmark \_\_\_\_\_

## Before collecting, I predict:

☐ No ☐ A few ☐ Some ☐ Many  
visible fragments or fibers, because ...

## Draw or mark the collection area:



Weather or visible conditions \_\_\_\_\_

Possible contamination or unusual condition \_\_\_\_\_

☐ Jar sealed ☐ Jar + lid labels match ☐ Site recorded

Record what you observe. A visible item is suspected, not confirmed, plastic.

CUT HERE



# Charlotte Beach sample record

Real field sample | Card 2

Pair \_\_\_\_\_ Sample ID \_\_\_\_\_ Time \_\_\_\_\_

Site or nearby landmark \_\_\_\_\_

## Before collecting, I predict:

☐ No ☐ A few ☐ Some ☐ Many  
visible fragments or fibers, because ...

## Draw or mark the collection area:



Weather or visible conditions \_\_\_\_\_

Possible contamination or unusual condition \_\_\_\_\_

☐ Jar sealed ☐ Jar + lid labels match ☐ Site recorded

Record what you observe. A visible item is suspected, not confirmed, plastic.



# Collect a traceable sand sample

Six pairs use the same method. Every real sample stays separate and connected to one place.

|                                                                                                                                                                            |                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>1</div>  <h2>Choose</h2> <p>Go to your assigned safe site.</p>                       | <div>2</div>  <h2>Frame</h2> <p>Place the red square flat on the sand.</p>                    |
| <div>3</div>  <h2>Record</h2> <p>Write the ID, time, landmark, and prediction.</p>        | <div>4</div>  <h2>Scoop</h2> <p>Use a clean spoon.<br/>Collect the top 1/2 inch.</p>          |
| <div>5</div>  <h2>Seal</h2> <p>Put sand in its own jar.<br/>Close it before moving.</p> | <div>6</div>  <h2>Label + map</h2> <p>Match jar, lid, card, and site. Return to Aastha.</p> |

**STOP:** Do not add salt water at the beach. Do not combine samples. Do not call a visible fragment confirmed plastic.

**Choose a role:** sampler, labeler, mapper, photographer, materials monitor, seated coordinator, or partner support.

# Invisible Invaders labels

SET 1 OF 2 | CUT ON DASHED LINES

PREPARED MODEL

**Durand reference model**

PREPARED MODEL

**Ontario reference model**

PREPARED MODEL

**Mystery bird clue**

FICTIONAL PROP

**Model bird — not a real case**

STORAGE BIN

**Prepared models only**

STORAGE BIN

**Sealed field samples only**

STORAGE BIN

**Clean tools**

STORAGE BIN

**Used tools**

FIELD SAMPLE

**II- \_\_\_\_\_ - \_\_\_\_\_ - \_\_\_\_\_ - \_\_\_\_\_**

FIELD SAMPLE

**II- \_\_\_\_\_ - \_\_\_\_\_ - \_\_\_\_\_ - \_\_\_\_\_**

FIELD SAMPLE

**II- \_\_\_\_\_ - \_\_\_\_\_ - \_\_\_\_\_ - \_\_\_\_\_**

FIELD SAMPLE

**II- \_\_\_\_\_ - \_\_\_\_\_ - \_\_\_\_\_ - \_\_\_\_\_**



# Choose how you join

Different routes can show the same careful scientific thinking. You may change routes at any time.

|                          |                                 |                             |                            |
|--------------------------|---------------------------------|-----------------------------|----------------------------|
| <br><b>Point</b>         | <br><b>Draw or write</b>        | <br><b>Speak or dictate</b> | <br><b>Photograph</b>      |
| <br><b>Use a partner</b> | <br><b>Choose a seated role</b> | <br><b>Ask for a scribe</b> | <br><b>Pass and rejoin</b> |

|                                            |                                                               |                                                        |
|--------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------|
| <b>OBSERVATION</b><br><b>I noticed ...</b> | <b>POSSIBLE MEANING</b><br><b>I think this could mean ...</b> | <b>UNCERTAINTY</b><br><b>We still need to know ...</b> |
|--------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------|

**Careful language protects people and the science:** prepared model, proxy piece, real field sample, suspected particle, supported, possible, or still unknown.



# Field Friday: the 40-minute routine

Use this at the station. It contains the complete sequence, the protected decisions, and the reset steps.

**ANCHOR:** Humans aren't trash cans, but we have plastics in our bodies.

**FRIDAY QUESTION:** How can plastic move among or collect in sand, water, air, living things, and human systems even when we do not notice it at first?

|       |                                 |                                                      |
|-------|---------------------------------|------------------------------------------------------|
| 0-3   | <b>Welcome + access preview</b> | Show the whole sequence. Offer participation routes. |
| 3-7   | <b>Fictional case</b>           | Name the story as a model immediately.               |
| 7-10  | <b>Demonstrate once</b>         | Predict → look → magnify → record → reset.           |
| 10-18 | <b>Durand model</b>             | One direction at a time. Record exact observations.  |
| 18-26 | <b>Ontario model</b>            | Repeat the same method. Keep pieces contained.       |
| 26-30 | <b>Match + reflect</b>          | Require two details and one model limit.             |
| 30-34 | <b>Choose sites</b>             | Select six safe, useful comparison locations.        |
| 34-38 | <b>Collect real sand</b>        | Frame → scoop → seal → label → map.                  |
| 38-40 | <b>Wonder + close</b>           | Save exact youth questions for Monday.               |

**Piter:** agenda, fictional-case disclosure, demonstration, claim-evidence-limit discussion, and close.

**Aastha:** materials, timing, proactive support, exact youth language, labels, maps, and reset checks.



# Protect these decisions

When time or energy drops, reduce repetition—not access, evidence limits, or traceability.



## BEFORE CAMPERS ARRIVE

- Pilot two reference models and the matching mystery clue.
- Record the known answer and exact proxy counts.
- Place model materials, field jars, clean tools, and used tools in separate bins.
- Prelabel six pair jars and two backups.
- Confirm safe boundaries, timing, weather, and photo permissions.



## AFTER EACH ROTATION

- Recover and count every prepared proxy piece.
- Reset the known model arrangement.
- Match field jar, lid, card, and map.
- Keep prepared models and real samples separate.
- Save exact youth questions and one access change.



## SAY THIS CLEARLY

**Case disclosure:** "This is a model story, not a report of a real injured bird."

**Evidence limit:** "A visible field piece is suspected, not confirmed, plastic."

**Close:** "The model had a known answer. The beach does not."



## NEVER CLAIM

- The reference models measure either beach.
- The fictional case explains animal illness.
- A visible field fragment is confirmed plastic.
- One observation proves a source, route, dose, or health effect.
- Young people or families are responsible for systems they do not control.

**Field stop:** No salt water at the beach. Never combine pair samples. Monday processing is per sample: SAMPLE → ADD BRINE → SETTLE → TRANSFER TOP LAYER → FILTER → COVER/DRY → INSPECT.